

Transformers and Large Language Models

DSC 340 Applied Machine Learning

Special Topics

Friday, May 2, 2025

GPT

- **Generative**
 - Generate coherent, contextually appropriate responses
- **Pretrained**
 - Trained on massive datasets to predict the next word effectively
- **Transformer**
 - Uses transformer architecture, effective at long-range dependencies

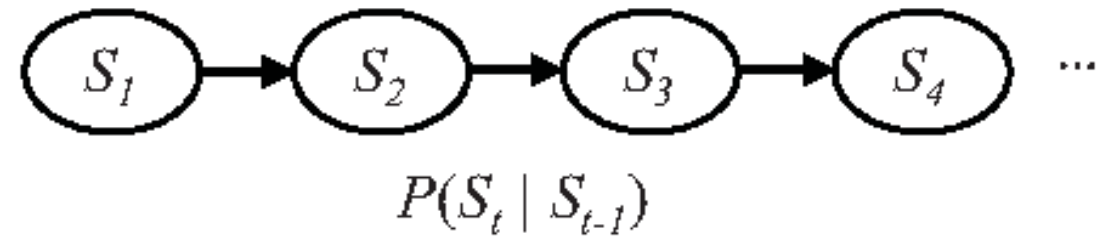
Why Study GPT and Transformers?

- Practical: Powering chatbots, translation, text-to-image, and more
- Theoretical: A leap from traditional neural networks
- Cultural: Ethical, ecological, and societal impacts

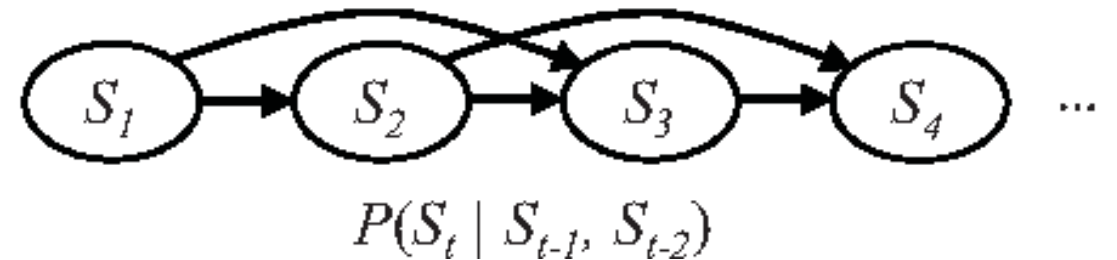
Language Models Recap

- Goal: Predict next word (token)
 “The dog chased the _____”
- Simple language models
 - Bag of words
 - First-order Markov chain
 - Second-order Markov chain
- Context matters:
 Long-range dependencies

a) Order 1 Markov Chain



b) Order 2 Markov Chain



Language Models → Probability Distribution

- In machine learning, language models are based on probabilities
 - Recall: Distribution learned data (corpus)
- Example: “The dog chased the _____”
 - Stop and Think: What is your prediction?
- Possible predictions (completely made up):

Ball:	30%
Squirrel:	25%
Mailman:	20%
Car:	20%
Neighbor:	3%

Tokens and Tokenization

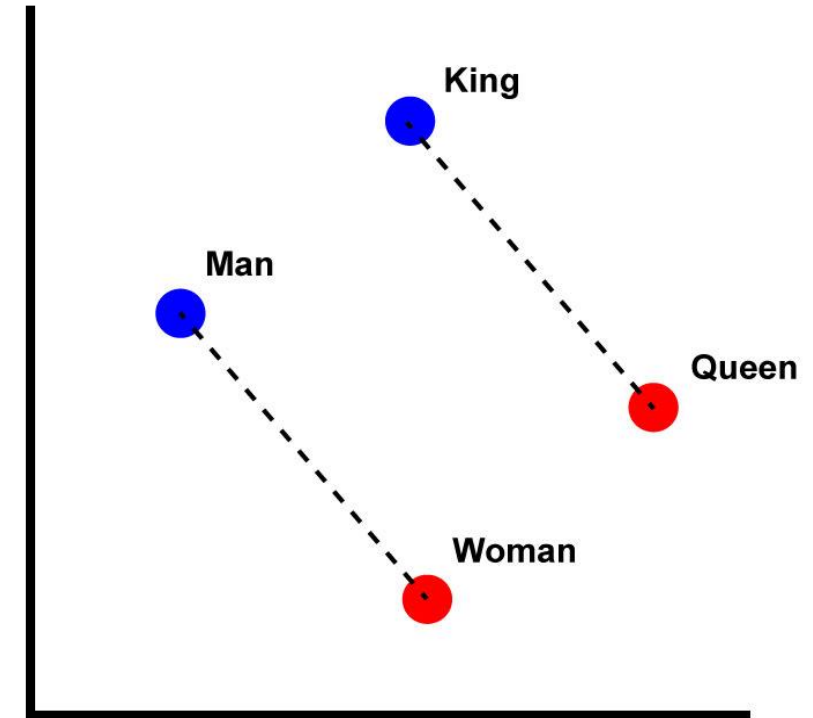
- Breaking text into **tokens** (e.g., words or **subwords**)
- Tokens sit between **letters** and **words** in granularity
- Example: “Out walking!” → [“out”, “walk”, “ing”, “!”]
- Smaller vocabulary: GPT-3 uses ~50,000 tokens vs. ~1M words
- Tokens often relate to **morphemes**, encoding meaning

Tokens and Model Behavior

- Tokens drive model predictions, not words
- Counting tokens $\neq$ Counting words
 - This is why even models like GPT-4o struggle with precise counts!
 - Example: Asking for exactly 10 words may fail
- Context window: Models “remember” a limited number of tokens

From Tokens to Embeddings

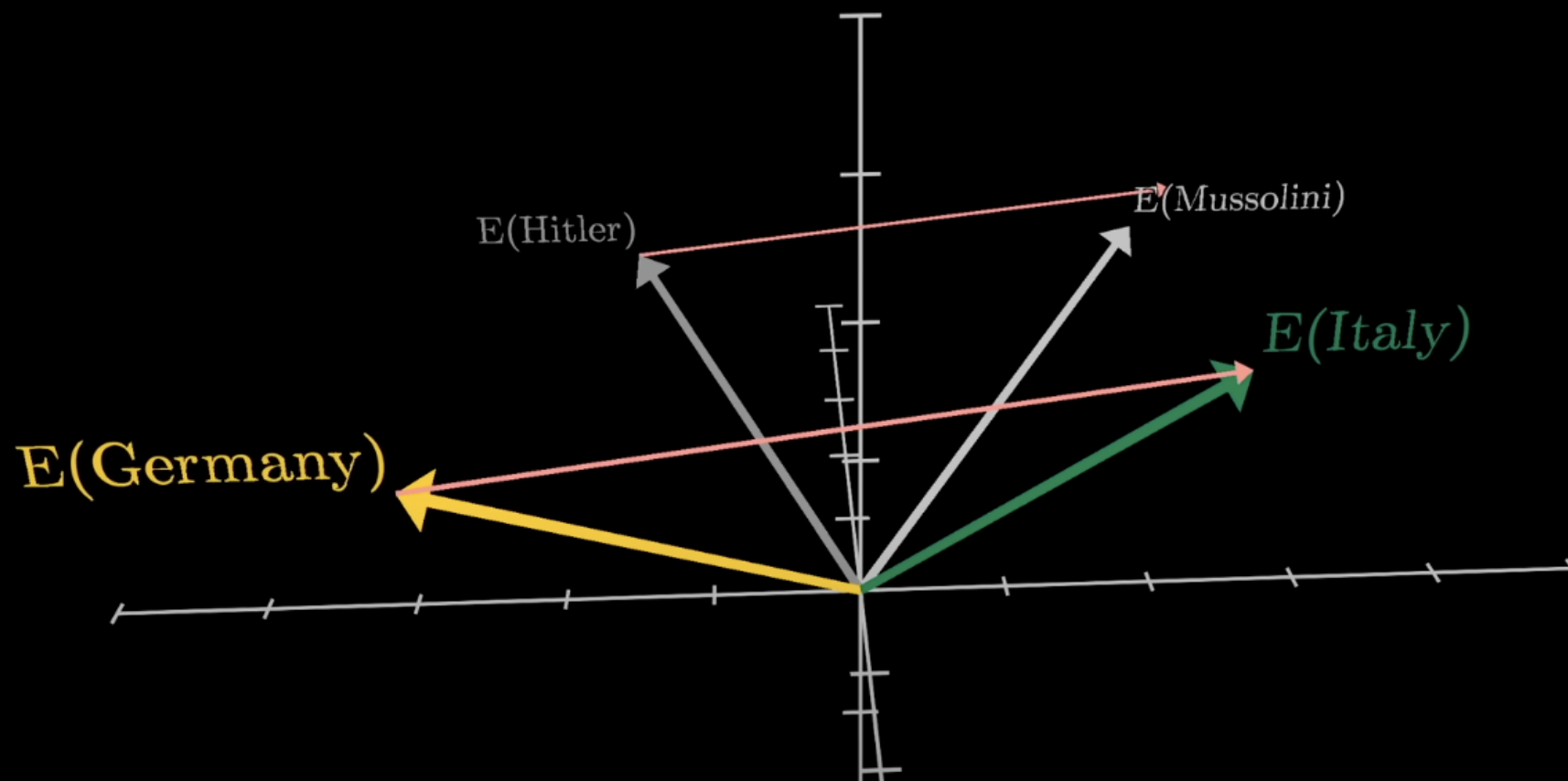
- Tokens become **vectors** in a high-dimensional space
- Embeddings encode **meaning** and **relationships**
- Example:
 $\text{"king"} - \text{"man"} + \text{"woman"} \approx \text{"queen"}$
- Word embeddings cluster similar words together



Understanding Word Embeddings

- Words with similar meanings are **closer** in the embedding space
- Contextual embeddings: Meaning changes with usage (e.g., “bank”)
- Pretrained embeddings save time and improve accuracy
- Applications: Search engines, translation, chatbots

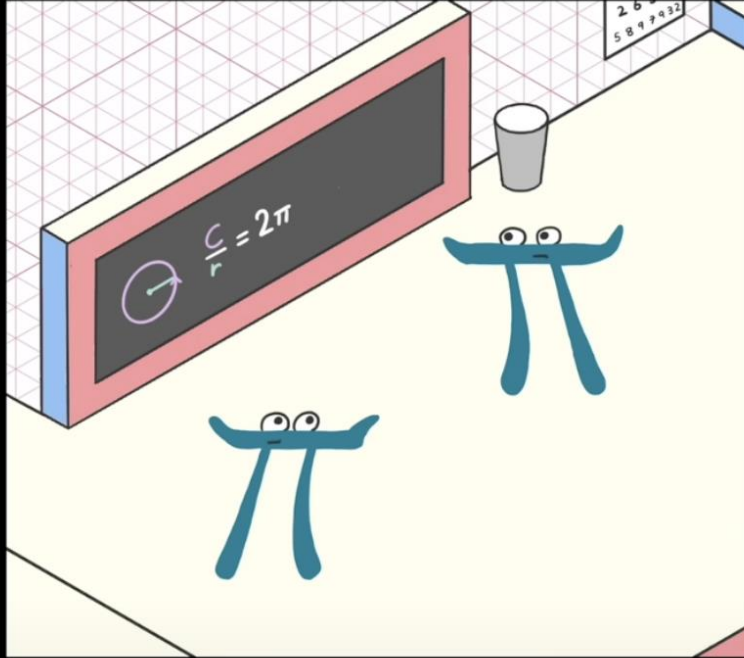
$$E(\text{Hitler}) + E(\text{Italy}) - E(\text{Germany}) \approx E(\text{Mussolini})$$



Generating Text: The Basics

- Generating Text: The Basics
- Task: Predict the **next token** based on context
- **Softmax**: Converts outputs into **probabilities**
 - Sigmoid : Step Function :: Softmax : Max
 - Forces all values to be between 0.0 and 1.0
 - Forces the values to add up to 1.0
- **Temperature**: Adjusts "creativity" in predictions
 - Low temperature → Predictable
 - High temperature → Surprising

Behold, a wild pi creature,
foraging in its native _____



land	22%
forest	9%
country	5%
habitat	4%
forests	4%
soil	4%
territory	2%
woods	2%
lands	1%
waters	1%
woodland	1%
grass	1%
⋮	



Controlling Text Generation

- Top-k sampling: Limits predictions to the **k most probable tokens**
- Top-p (nucleus) sampling: Focuses on a **dynamic probability threshold**
- Temperature, Top-k, and Top-p shape creativity and coherence
- Applications: Chatbots, story generation, creative writing

Challenges with Long-Range Dependencies

- Older models: Struggled with distant relationships

"The cat ate its food because **it** was hungry."

- Ambiguity: Does **it** refer **food**? **cat**? A **dog** mentioned earlier?
- Limited context: Older models had small **context windows**

Short distance

One mole of carbon dioxide

Long distance

Harry Potter was a highly unusual boy in many ways. For one thing, he hated the summer holidays more than any other time of year. For another, he really wanted to do his homework but was forced to do it in secret, in the dead of night. And he also happened to be a wizard.

It was nearly midnight, and he was lying on his stomach in bed, the blankets drawn right over his head like a tent, a flashlight in one hand and a large leather-bound book (A History of Magic by Bathilda Bagshot) propped open against the pillow. Harry moved the tip of his eagle-feather quill down the page, frowning as he looked for something that would help him write his essay, "Witch Burning in the Fourteenth Century Was Completely Pointless discuss."

The quill paused at the top of a likely-looking paragraph. Harry Pushed his round glasses up the bridge of his nose, moved his flashlight closer to the book, and read:

Non-magic people (more commonly known as Muggles) were particularly afraid of magic in medieval times, but not very good at recognizing it. On the rare occasion that they did catch a real witch or wizard, burning had no effect whatsoever. The witch or wizard would perform a basic Flame Freezing Charm and then pretend to shriek with pain while enjoying a gentle, tickling sensation. Indeed, Wendelin the Weird enjoyed being burned so much that she allowed herself to be caught no less than fortyseven times in various disguises.

Harry put his quill between his teeth and reached underneath his pillow for his ink bottle and a roll of parchment. Slowly and very carefully he unscrewed the ink bottle, dipped his quill into it, and began to write, pausing every now and then to listen, because if any of the Dursleys heard the scratching of his quill on their way to the bathroom, he'd probably find himself locked in the cupboard under the stairs for the rest of the summer.

The Dursley family of number four, Privet Drive, was the reason that Harry never enjoyed his summer holidays. Uncle Vernon, Aunt Petunia, and their son, Dudley, were Harry's only living relatives. They were Muggles, and they had a very medieval attitude toward magic. Harry's dead parents, who had been a witch and wizard themselves, were never mentioned under the Dursleys' roof. For years, Aunt Petunia and Uncle Vernon had hoped that if they kept Harry as downtrodden as possible, they would be able to squash the magic out of him. To their fury, they had been unsuccessful. These days they lived in terror of anyone finding out that Harry had spent most of the last two years at Hogwarts School of Witchcraft and Wizardry. The most they could do, however, was to lock away Harry's spellbooks, wand, cauldron, and broomstick at the start of the summer break, and forbid him to talk to the neighbors.

This separation from his spellbooks had been a real problem for Harry, because his teachers at Hogwarts had given him a lot of holiday work. One of the essays, a particularly nasty one about shrinking potions, was for Harry's

How Transformers Address Context

- Attention mechanism: Weighs **relevant parts of input** text
- Example: “The cat ate its food because it was hungry.”
- Self-attention: Focuses on “cat” for **it was hungry**
- Large context windows:
Modern models can process **thousands of tokens**

Transformer Architecture Overview

- Stacked layers
 - Combine **attention** and **feed-forward** components
- Residual connections
 - Preserve information across layers
- Layer normalization
 - Stabilizes learning for better performance
- Why stacking matters
 - Deeper models learn more complex patterns

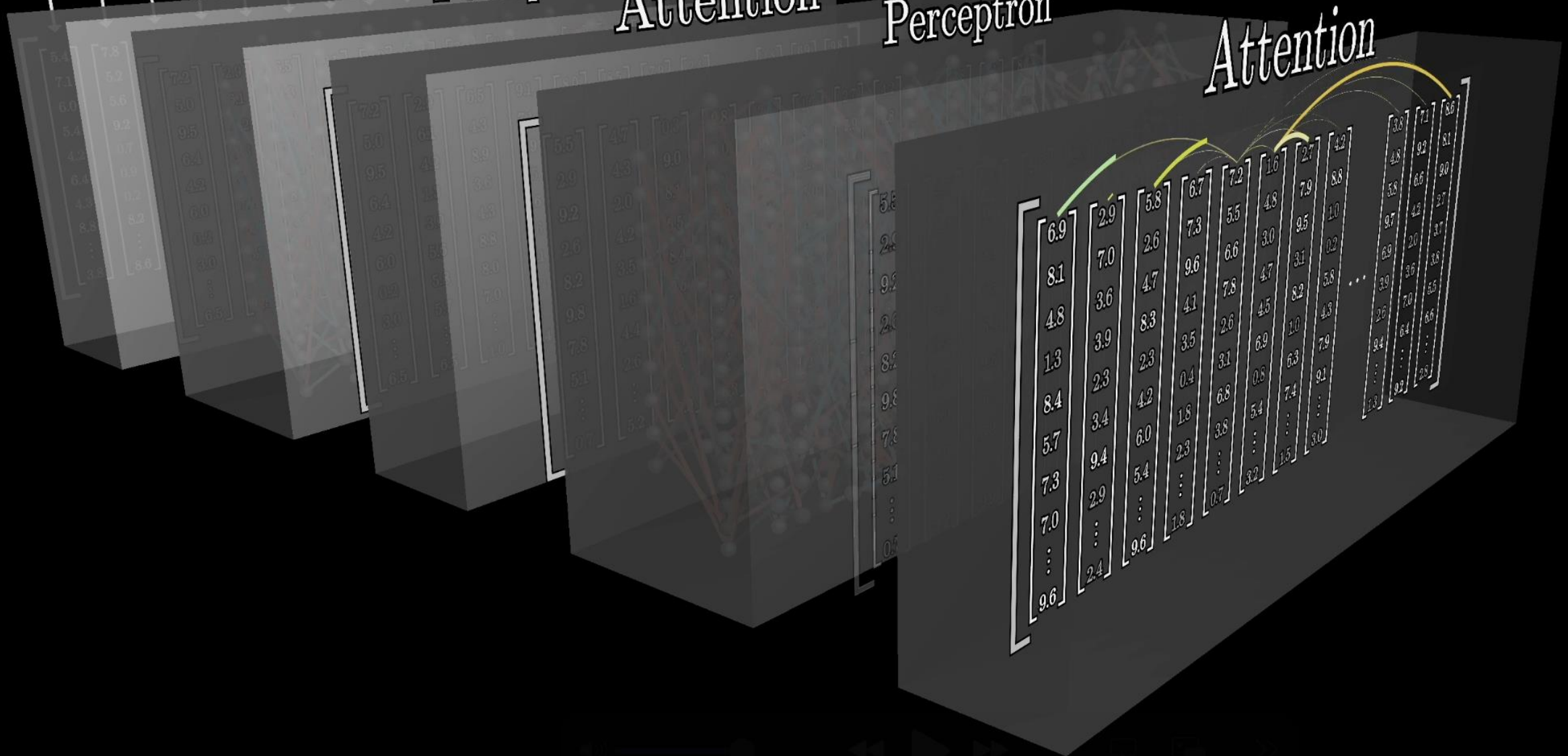
To date the clearest thinker of

Attention

Perceptron

Multilayer
Perceptron

Attention



Key Components of Transformer Layers

- Attention mechanism
 - Focuses on **relevant tokens** in input
- Feed-forward network
 - Multilayer perceptron (MLP)
 - Processes information **independently** for each token
- Residual connections
 - Prevent information loss during processing
- Layer normalization
 - Keeps outputs **stable and efficient**

Attention Mechanism: The Basics

- Key idea: Focus on **important words** in the input
- Attention scores: Measure **relevance** between tokens
- Example: “The cat ate because it was hungry.”
- Focus: Link “it” to “cat,” not “ate”

How Attention Works: Queries, Keys, Values

Each token creates:

- **Query:** What are we looking for?
- **Key:** What information do we have?
- **Value:** The actual information

Example: Linking “it” to “cat”

- **Query:** “it” → What is “it” referring to?
- **Key:** “cat” → How relevant is “cat” to “it”?
- **Value:** “cat” → The actual meaning to focus on

Self-Attention: Key Feature of Transformers

- Self-attention: Each token attends to **all tokens**, including itself
- Importance: Captures **relationships across the sequence**
- Efficient: Parallel processing speeds up computation

Multi-Headed Attention: The Basics

- Why multiple heads?
 - Focus on **different aspects** of relationships in the text
 - Example: Syntax vs. semantics
- How does it work?
 - Split queries, keys, and values into multiple **heads**
 - Process each head independently
- Combine results for richer understanding

Why Multi-Headed Attention Matters

- Handles complex relationships: Syntax, semantics, and context
- Enables parallel processing for efficiency
- Example: “The cat chased the mouse because it was hungry.”
 - Head 1: Links “it” to “cat” (semantic focus)
 - Head 2: Tracks subject-verb agreement (syntax focus)

How Multi-Headed Attention Works

- Split: Divide queries, keys, and values into smaller subsets
- Independent processing: Each head focuses on a unique aspect
- Combine: Merge results into a single output

Positional Encoding

- Transformers process inputs in parallel
- Need to encode **order** of tokens
- Example: Adding positional embeddings

Fine-Tuning: Adapting Pretrained Models

- Process: Customize a general model for a specific task
- Saves time and resources: Build on pre-trained knowledge
- Example: Fine-tune GPT for medical Q&A or legal analysis

RLHF

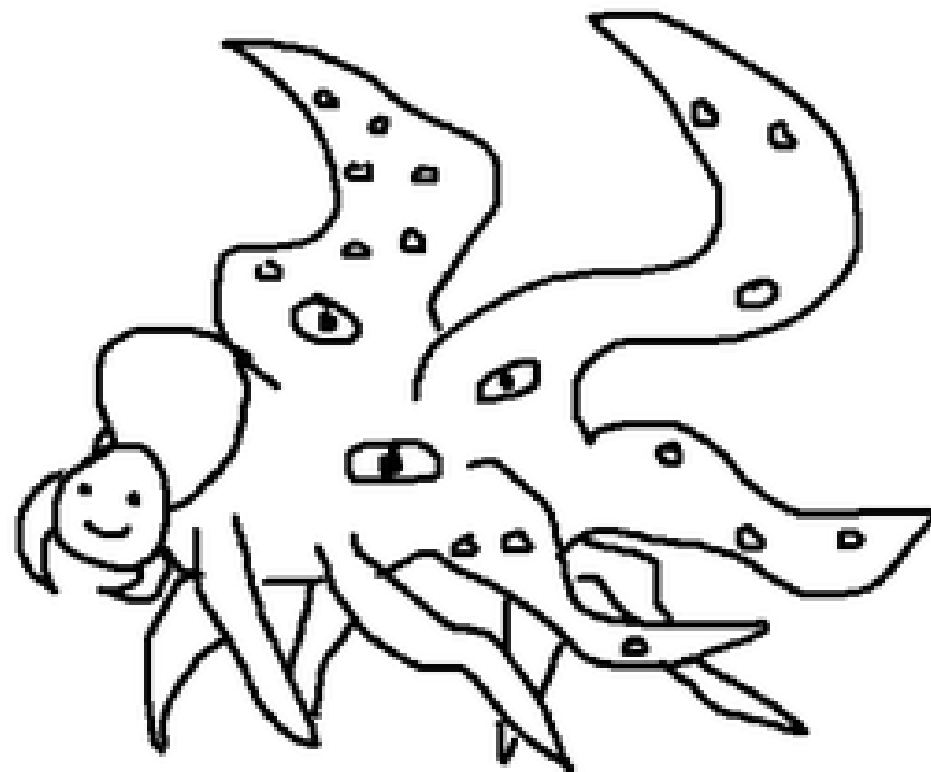
Reinforcement Learning with Human Feedback

- Aligning Models with Human Intent
- Combines **human input** with reinforcement learning
- Human feedback guides the model's behavior
- Example: Training ChatGPT to provide polite, helpful answers

GPT-3



GPT-3 + RLHF



Shoggoth RLHF Meme

<https://www.nytimes.com/2023/05/30/technology/shoggoth-meme-ai.html>

Applications

- Chatbots (e.g., customer service, tutoring)
- Translation (e.g., real-time, multilingual systems)
- Creative writing (e.g., poetry, screenwriting, brainstorming)

Limitations and Concerns

- **Context window:** Limited capacity for long documents
- **Hallucinations:** Generates false or misleading information
- **Biases:** Reflects biases in training data
- **Ethical concerns:** Energy consumption, misinformation risks

Wrap-Up and Next Steps

- Transformers revolutionized AI with **attention**
- GPT is a **powerful generative model**
- Next time: **LLMs in agents and RAG**